P739 mathematical workload gate: finite accounting observations

30 October 2026

This report records observations only, with no scientific verdict or general ranking. It covers public synthetic known-answer mathematical workloads in run `pencil-p739-20261030-a1`, the prescribed 60-target $p = 739$ gate. It is not the full 1,260-target campaign.

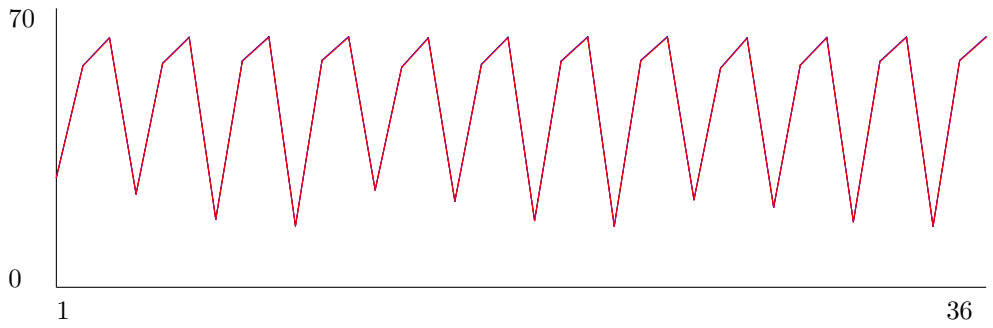
Recorded scope. All 1,080 planned executions are present: 180 warmups and 900 measured executions over candidate, rho, and rho A/A controls. Each arm has 360 verified outcomes including warmups. The finite-cost comparison uses all 60 targets and all five measured streams, yielding 300 candidate/reference pairs per cell. Every one of the 36 cells admits all 300 pairs. Warmups remain in the observations and are excluded from measured cost sums.

Derived counted costs. Ratios divide summed candidate cost by summed rho cost across the same 300 pairs; they are not means of per-target ratios. Method-only ratios span 16.424524–67.416795; including calibration once spans 16.382345–67.242171 (rounded display). At $S/M = 0.5$, $I/M = 10$, native charge 0.1 GAE/event, exact method/rho is 10535665436/175639669 and exact method-plus-calibration/rho is 10536131936/176106169. The full exact grid is supplied, including every cell; no outcomes are discarded to form these sums.

Finite model and exclusions. The declared model counts field/scalar arithmetic, constructor validation, setup/transfer, collection/graph/elimination, hash/RNG, source verification, metadata clone, and calibration. Primitive events are disjoint from diagnostic aggregate counters. Reporting, allocator internals, observer overhead, process/harness/audit work, infrastructure, and retries remain separate resources with no invented GAE conversion. This is neither a CPU-instruction census nor a universal correctness proof. The post-data binding review reuses the pre-data independent source audit and exact pinned instrument bytes; it does not relabel post-data work as preregistration.

Separate resources and control observations. The controller receipt records start 00:15:44.603 UTC and end 00:26:26.317 UTC on 30 October; its terminal status is FAILED, with the controller command returning exit 1. The complete fresh observation set is retained separately from this orchestration status. The saved job status records start 00:21:50 UTC and failure at 00:26:10 UTC with BackoffLimitExceeded. These intervals include distinct operational scopes and are not candidate/rho wall-time ratios. Recorded isolation and A/A controls do not qualify a wall-time comparison: qualified ratios stay null. An earlier attempt failed before payload PID assignment with EACCES because the progress executable was mode 0700; its failed attempt and the metadata-only correction are retained. The prior 180-second reservation is not measured runtime. No cost conversion or subtraction conceals that failure.

Identity and review files. The instrument is commit `543a54b4...`, binary `a1c8ca89...`; frozen version5 protocol body digest is `cf8f76ea...`. The exact raw-record, protocol, analysis and source-review digests are in observations.json and linked receipts. The source record is validation/covered_gate_results_20261030/covered-analysis-receipt.json; the raw versioned locators and separate controller/job resources are in validation/actual_gate_results_20261030. Full method, seeds, caps, workload identities and all observed records remain bound by those manifests. This renderer only transforms existing exact values for display; it launches no workload.



Blue: method only. Red: method plus calibration. Horizontal: price-cell index; vertical: candidate/rho ratio. Graph shows all 36 recorded price cells, each using all 300 measured pairs. Points are derived from exact rational output; display rounding only. There are no statistical error bars: the grid varies declared prices, not a confidence interval.

All recorded price cells

Every row admits 300 of 300 pairs with zero missing/ineligible pairs. S and I are prices relative to M; N is GAE per declared native event. Exact numerator/denominator pairs are in cost-grid-exact.csv and observations.json. Ratios below are rounded to six decimals for display.

Cell	S/M	I/M	N	Method/rho	With calibration/rho
1	0.0	1	0.0	29.555960	29.477422
2	0.0	1	0.1	59.622508	59.466872
3	0.0	1	1.0	67.161009	66.986943
4	0.0	10	0.0	25.106901	25.040729
5	0.0	10	0.1	60.268188	60.111173
6	0.0	10	1.0	67.282603	67.108271
7	0.0	100	0.0	18.296137	18.248802
8	0.0	100	0.1	60.918606	60.760209
9	0.0	100	1.0	67.395416	67.220838
10	0.0	1000	0.0	16.494841	16.452470
11	0.0	1000	0.1	61.047968	60.889297
12	0.0	1000	1.0	67.416795	67.242171
13	0.5	1	0.0	26.196410	26.127139
14	0.5	1	0.1	59.246336	59.091776
15	0.5	1	1.0	67.125056	66.951098
16	0.5	10	0.0	23.153682	23.092878
17	0.5	10	0.1	59.984544	59.828296
18	0.5	10	1.0	67.250409	67.076164
19	0.5	100	0.0	17.957584	17.911174
20	0.5	100	0.1	60.850576	60.692356
21	0.5	100	1.0	67.387256	67.212699
22	0.5	1000	0.0	16.459597	16.417321
23	0.5	1000	0.1	61.040165	60.881515
24	0.5	1000	1.0	67.415856	67.241234
25	1.0	1	0.0	23.569046	23.507018
26	1.0	1	0.1	58.972354	58.818576
27	1.0	1	1.0	67.099313	66.925432
28	1.0	10	0.0	21.505981	21.449705
29	1.0	10	0.1	59.749328	59.593715
30	1.0	10	1.0	67.223802	67.049629
31	1.0	100	0.0	17.632883	17.587360
32	1.0	100	0.1	60.785010	60.626960
33	1.0	100	1.0	67.379385	67.204849
34	1.0	1000	0.0	16.424524	16.382345
35	1.0	1000	0.1	61.032394	60.873764
36	1.0	1000	1.0	67.414921	67.240301

The JSON retains exact summed candidate and reference costs, not just ratios. The CSV preserves all 36 exact ratio fractions. Earlier null-coverage observations remain separate and unchanged.